Bayesian discrete trait analysis

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PhylogeogRaphy Workshop

Day 4

1. Quick recap
2. Joint estimation of trees and character change
3. Bayesian inference BEAST2

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A quick recap so far:

We've learned about phylogeographic methods that treat a phylogeny as given:

- discrete trait analysis
  - ace - obtain joint and/or marginal reconstructions at internal nodes
  - simmap - sample a joint reconstruction at nodes, and simulate state changes along edges
  - SSE (BiSSE, MuSSE, etc) - obtain joint and/or marginal reconstructions at internal nodes while accounting for correlations between birth/death rates and state
  - SAASI - extend SSE models to accommodate different sampling rates by lineage

These have all assumed we have a fixed tree we can depend on. It may be preferable to account for phylogenetic uncertainty.

A quick recap so far:

It is typical in the field to jointly estimate parameters of phylodynamics models jointly with a phylogeny. We will explore this approach in the simplest case (basic birth-death models that treat all lineages as identical).

Then, we will extend this in two ways:

1. multi-type birth death models (this afternoon)
2. Bayesian stochastic search variable selection (tomorrow)

Sorry, structured coalescent fans.

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Character evolution likelihood

For ancestral character estimation and stochastic character mapping, we considered fixed trees, $\mathcal{T}$, and defined Markov chains on them to obtain a likelihood for tip states, $\mathbf{x}$,

$$p(\mathbf{x}|\mathcal{T}) = \sum_{\mathbf{y}} p(\mathbf{x}, \mathbf{y}|\mathcal{T}),$$

where we are marginalizing over unobserved configurations of node states, $\mathbf{y}$

Nucleotide evolution evolution likelihood

We have not discussed phylogenetic reconstruction in this workshop, but at this point we probably should.

We have used $\mathbf{x}$ to represent tip states of interest (geography or traits). Let's introduce $\mathbf{s}$ to represent our multiple sequence alignment.

Nucleotide substitution models use Markov chains just like the trait evolution models we saw in ace and simmap. 4 nucleotides means 4 states of the Markov chains.

If we let $\mathbf{u}$ represent the unobserved configuration of ancestral nucleotide states, the sequence likelihood given the tree has the form

$$p(\mathbf{s}|\mathcal{T}) = \sum_{\mathbf{u}} p(\mathbf{s}, \mathbf{u}|\mathcal{T}).$$

Nucleotide evolution evolution likelihood

This

$$p(\mathbf{s}|\mathcal{T}) = \sum_{\mathbf{u}} p(\mathbf{s}, \mathbf{u}|\mathcal{T}),$$

is exactly the form of our likelihoods up to now, except that the states correspond to nucleotides in the sequence alignment.

With a large collection of sites in the genome, we usually calculate likelihoods at each site and multiply them all together to obtain the likelihood of the sequences:

$$\begin{aligned} p(\mathbf{s}|\mathcal{T}) &= \prod_i p(s_i|\mathcal{T}), \\ &= \prod_i \sum_{\mathbf{u}} p(s_i, \mathbf{u}_i|\mathcal{T}). \end{aligned}$$

Sequence and character likelihood

Maximum Likelihood phylogenetic reconstruction maximizes $p(\mathbf{s}|\mathcal{T})$ over trees $\mathcal{T}$ and substitution rate parameters (mathematically like the q_{ij}'s in our models so far)

A Bayesian approach adopts a slightly different approach, by **sampling** from the posterior distribution

$$p(\mathcal{T}|\mathbf{s}) \propto p(\mathbf{s}|\mathcal{T})p(\mathcal{T}).$$

which requires specification of a "tree prior", $p(\mathcal{T})$.

The prior $p(\mathcal{T})$ is some model for generating trees, such as the Yule model, birth-death model, etc. It could also be a coalescent model (which we have not, and will not discuss for time's sake)

Sequence and character likelihood

After sampling a distribution of trees from

$$p(\mathcal{T}|\mathbf{s}) \propto p(\mathbf{s}|\mathcal{T})p(\mathcal{T}).$$

we could use ace, simmap, or SSE models to estimate ancestral states.

In the Bayesian setting, researchers typically estimate everything together by considering the **joint likelihood**

$$p(\mathbf{s},\mathbf{x}|\mathcal{T}) = \sum_{\mathbf{u},\mathbf{y}} p(\mathbf{s},\mathbf{x},\mathbf{u},\mathbf{y}|\mathcal{T}),$$

that is, by simultaneously estimating trees and characters at nodes/edges.

Joint character and sequence likelihood

The simplest version of this approach models trait evolution independently of sequences, conditional on a tree, but nonetheless jointly with the estimation of the phylogeny[1]:

$$
\begin{aligned}
p(\mathbf{s}, \mathbf{x} | \mathcal{T}) &= \sum_{\mathbf{u},\mathbf{y}} p(\mathbf{s}, \mathbf{x}, \mathbf{u}, \mathbf{y} | \mathcal{T}), \\
&= \sum_{\mathbf{u},\mathbf{y}} p(\mathbf{s}, \mathbf{u} | \mathcal{T}) p(\mathbf{x}, \mathbf{y} | \mathcal{T}), \\
&= \sum_{\mathbf{u}} p(\mathbf{s}, \mathbf{u} | \mathcal{T}) \sum_{\mathbf{u}} p(\mathbf{x}, \mathbf{y} | \mathcal{T})
\end{aligned}
$$

Note: the $p(\mathbf{x}, \mathbf{y} | \mathcal{T})$ depends on the generator matrix $\mathbf{Q}$ of our geography model, as well as substitution rates for the nucleotides (but I'm suppressing extra notation for readability)

[1] Philippe Lemey et al. "Bayesian phylogeography finds its roots". In: *PLoS computational biology* 5.9 (2009), e1000520.

Joint character and sequence likelihood

Taking a step back to examine this:

$$\begin{aligned} p(\mathbf{s}, \mathbf{x} | \mathcal{T}) &= \sum_{\mathbf{u},\mathbf{y}} p(\mathbf{s}, \mathbf{x}, \mathbf{u}, \mathbf{y} | \mathcal{T}), \\ &= \sum_{\mathbf{u},\mathbf{y}} p(\mathbf{s}, \mathbf{u} | \mathcal{T}) p(\mathbf{x}, \mathbf{y} | \mathcal{T}), \\ &= \sum_{\mathbf{u}} p(\mathbf{s}, \mathbf{u} | \mathcal{T}) \sum_{\mathbf{u}} p(\mathbf{x}, \mathbf{y} | \mathcal{T}), \end{aligned}$$

we see that we could estimate $\mathcal{T}$ with Maximum Likelihood, meaning that we use of information contained in both s and x to reconstruct the phylogeny.

Question How sensitive is our estimated phylogeny, $\hat{\mathcal{T}}$ to inclusion of tip states x (particularly if we have lots of sequence data)?

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Bayesian inference in BEAST2

For now, we will just jointly estimate everything to learn about this functionality in BEAST2.

We are focusing on the situation where we can treat character evolution independent of sequence evolution, given a tree,

$$\begin{aligned} p(\mathbf{s}, \mathbf{x}|\theta) &= \int_{\mathcal{T}} \sum_{\mathbf{u},\mathbf{y}} p(\mathbf{s}, \mathbf{x}, \mathbf{u}, \mathbf{y}|\mathcal{T}) p(\mathcal{T}|\theta), \\ &= \int_{\mathcal{T}} \sum_{\mathbf{u}} p(\mathbf{s}, \mathbf{u}|\mathcal{T}) \sum_{\mathbf{y}} p(\mathbf{x}, \mathbf{y}|\mathcal{T}) p(\mathcal{T}|\theta) \end{aligned}$$

Bayesian inference in BEAST2

A Bayesian analysis in BEAST2 uses MCMC to sample a collection of trees and parameters. We supply a prior distribution $p(\theta)$ for the tree-generating model (birth-death, coalescent, whatever), as well as a prior distribution $p(\mathbf{Q})$ for the character evolution model, as well as $p(\mathbf{A})$ for the nucleotide evolution model:

$$p(\theta, \mathbf{Q}, \mathbf{A} | \mathbf{s}, \mathbf{x}) \propto \int_{\mathcal{T}} \sum_{\mathbf{u}} p(\mathbf{s}, \mathbf{u} | \mathcal{T}, \mathbf{A}) \sum_{\mathbf{y}} p(\mathbf{x}, \mathbf{y} | \mathcal{T}, \mathbf{Q}) p(\mathcal{T} | \theta) p(\theta) p(\mathbf{Q}) p(\mathbf{A})$$

Bayesian inference in BEAST2

The multi-type birth death model, the structured coalescent model, and their variants deal with the situation where sequence evolution and character change are **not** conditionally independent given the tree:

$$p(\theta, \mathbf{Q}, \mathbf{A} | \mathbf{s}, \mathbf{x}) \propto \int_{\mathcal{T}} \sum_{\mathbf{u},\mathbf{y}} p(\mathbf{s}, \mathbf{x}, \mathbf{u}, \mathbf{y} | \mathcal{T}) p(\mathcal{T} | \theta) p(\theta) p(\mathbf{Q}) p(\mathbf{A})$$

Plan for the afternoon and tomorrow

We will carry out a few analyses on the same data to see how things change:

1. (Today) Estimate influenza phylogenies in BEAST2, without supplying geographic location. Then, fit ace/fitMK, simmap to a sample of trees from the posterior collection.
2. (Today) Estimate influenza phylogenies in BEAST2 jointly with geographic location, using the same model for trait change as in the previous step
3. (Tomorrow) Repeat the previous step but using Bayesian Stochastic Search Variable Selection (BSSVS) to obtain regularized estimates for migration rates. We can compare this to ace+LASSO from the trees we get in Step 1.